

allowing semiconductors to play critical roles in computing, appliances, communication devices, transportation systems, and critical infrastructure. The 2nm design demonstrates the development of more transistors on a chip. This means that

processor designers now have more options to infuse core-level innovations for improving capabilities of leading edge workloads like AI and cloud computing, as well as provide new pathways for hardware-enforced security and encryption.

New material could lead to devices that use very little energy



Harnessing the hum of fluorescent lights for more efficient computing (Credit: <https://news.umich.edu>)

A team led by the University of Michigan researchers has developed a material that is at least twice as ‘magnetostrictive’ and costs less. In addition to computing, it could also lead to better magnetic sensors for medical and security devices. Magnetostriction occurs when a material’s shape and magnetic field are linked, that is, a change in shape causes a change in magnetic field. The property could be key to a new generation of computing devices called magnetoelectrics. Magnetoelectric devices use magnetic fields instead of electricity to store the digital ones and zeros of binary data. Tiny pulses of electricity cause them to expand or contract slightly, flipping their magnetic field from positive to negative or vice versa. And because they don’t require a steady stream of electricity, they can effectively work by simply using very little energy.

New system that makes MRI simple for infants

Newborns and infants can have a variety of complications in their first months and years of life. To prevent long-term disorder, possible damage must be diagnosed as early as possible. The common imaging methods for diagnosing pediatric diseases are ultrasound and x-rays. Although MRI offers high-resolution imaging of all organs and helps reveal many pediatric problems, MRI examinations usually require sedation of the newborn, which can be risky. Therefore, Neoscan Solutions in collaboration with Otto von Guericke University Magdeburg has developed a new MRI system especially for newborns and infants, expanding their diagnostic options that makes MRI simple and available whenever and wherever it is needed.



MRI as individual as newborns and infants need it (Credit: www.neoscan-solutions.com)